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Modeling of lightcurves of binary asteroids

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ABSTRACT

We present a numerical method for inverting long-period components of lightcurves of asynchronous binary asteroids. Data of five near-Earth binary asteroids, (175706) 1996 FG_3, (65803) Didymos, (66391) 1999 KW_4, (185851) 2000 DP_{107} and (66063) 1998 RO_1, for two of them from more than one apparition, were inverted. Their mutual orbits' poles and Keplerian elements, size ratios, and ellipsoidal shape axial ratios were estimated via this inversion. The pole solutions and size ratios for 1999 KW_4 and 2000 DP_{107} are in a good agreement with independent estimates from radar measurements. We show that uncertainties of estimates of bulk densities of binary systems can be large, especially when observed on short arcs.

1. Introduction

In recent years a substantial progress has been made in discovery of binary systems in all asteroidal populations in the Solar System, and in characterization of their properties. They form a substantial part of near-Earth asteroids (NEA) as well as of small Main Belt asteroids. About 15% of NEAs larger than 0.3 km are binary systems, mostly asynchronous (Pravec et al., 2006), and a similar fraction of binaries is being observed among Main Belt asteroids (Pravec and Harris, 2007, and references therein).

After the discovery of asynchronous systems among near-Earth asteroids (NEAs) with doubly periodic lightcurves in the 1990s, the first successful attempts to model their lightcurves were made by Pravec et al. (2000) and Mottola and Lahulla (2000) in the case of NEA (175706) 1996 FG_3. Although the data and numerical implementation of their models differ, they obtained similar results, providing an independent verification. In the former work, trial coordinates of a mutual orbit's pole were used, but the pole location agrees within 10 degrees with a retrograde solution of the latter work, where a full-grid search of the orbital pole was carried out.

More extensive work on the topic was done by Pravec et al. (2006). Of the seventeen asynchronous binary NEAs whose data were analyzed, four (one in two apparitions) lightcurve datasets were inverted in a least-square sense to provide basic parameters of the systems.

The photometric observations are strongly supported by radar, and vice versa. The first radar detection of a binary system was NEA (185851) 2000 DP_{107} (Ostro et al., 2000; Margot et al., 2002), subsequently investigated by photometry (Pravec et al., 2006). Including this case the total number of binary systems observed by both methods is 15, thus validating the photometry as a reliable way to detect them. Since a strength of the radar echo is inversely dependent on the fourth power of a distance to the object, the radar technique is limited to close approaches of asteroids to the Earth. On the other hand, the radar imaging during favorable apparitions is able to obtain a very detailed information about shapes, spin states, and a dynamics of mutual orbit of the components. The best studied case using this method is NEA (66391) 1999 KW_4 (Ostro et al., 2006; Scheeres et al., 2006).

Knowledge of binaries' rotational and orbital properties is crucial for understanding their origin and evolution. Nowadays, the investigation of small binaries among NEAs is dominated by photometry and radar, in the Main Belt by photometry only. The main aim of this work is to describe an inversion method for obtaining parameters from photometric observations. A special emphasis was given to assess uncertainties of plausible solutions and to demonstrate limitations of the lightcurve inversion method.

Assumption of spherical shapes of components usually leads to bulk densities which are significantly off. The densities derived in this way should be therefore used with caution, in particular the uncertainty in density so derived generally does not allow a meaningful conclusion regarding composition or structure (porosity and macroscopic concavities).

2. Principles of photometric detection of asynchronous binaries

The standard reduction technique for the photometric data of binary asteroids has been developed and applied in previous works. We refer the reader to Pravec et al. (2006) for detailed description of the methods, and only their most important points are mentioned below.

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Photometric data of a binary asteroid, reduced to unit distances from Earth and Sun and to a given phase angle, in general, consist of three periodic components, which add linearly[1] into a combined lightcurve

$$F(t) = F_1(t) + F_2(t) + F_{\mathrm{mut}}(t), \tag{1}$$

where $F(t)$ is the total reduced flux at time t and $F_1(t)$ and $F_2(t)$ are rotational lightcurves produced by the two bodies of the binary system. $F_{\mathrm{mut}}(t)$ represents attenuations produced by occultations and/or eclipses, collectively called mutual events, that are superimposed onto the rotational lightcurves of the bodies.[2] Mutual events occur only in favorable geometric conditions when Earth or Sun are close to the mutual orbital plane of the system. Conventionally, F_1 refers to larger and F_2 to smaller body of the system. They are called primary and secondary, respectively.[3] Periods of the functions F_1 and F_2 (P_1 and P_2, respectively) are synodic rotational periods of the corresponding bodies. Primaries of most of photometrically observed binary asteroids rotate more rapidly than secondaries. The mutual component F_{mut} have a period $P_{\mathrm{orb}}^{\mathrm{syn}}$ that we define as synodic orbital period of the system. Strictly speaking, F_{mut} is not exactly a periodic function—the shape of mutual events may evolve rapidly with changing geometry of observations. However, a recurrence of the events allows a derivation of $P_{\mathrm{orb}}^{\mathrm{syn}}$ in most cases.

A binary system is called asynchronous if $P_1 \neq P_{\mathrm{orb}}$. Secondaries of asynchronous systems may or may not rotate synchronously with the orbital motion. If they do, then $P_2 = P_{\mathrm{orb}}^{\mathrm{syn}}$.

By fitting a two-period Fourier series (see Pravec et al., 2006) to data points outside mutual events, the rotational lightcurves $F_1(t)$ and $F_2(t)$ can be separated. In principle, the short period component can be used for modeling a shape of the primary. Kaasalainen et al. (2002) mentioned that three or four apparitions are usually required for modeling shapes of asteroids, while one or two apparitions are sufficient only in cases where the observing geometry varies considerably. Observations obtained for binary asteroids so far are not abundant enough and they do not satisfy the requirements. Instead of attempting to fit the primary's lightcurve by a shape that probably has nothing to do with real shape of the primary, we choose to subtract the short-period component $F_1(t)$ from the lightcurve. Remaining long-period component containing the mutual events and the secondary rotation lightcurve is then fitted in our numerical modeling. A rotationally symmetric shape is then used for the primary in simulations of mutual events.

The properties of the mutual events allows to estimate some other parameters. If the secondary is fully occulted or eclipsed, a flat bottom of constant brightness occur in the secondary event. For same albedos and phase effects of the two bodies, a depth of the flat bottom is related with a ratio of mean projected diameters of the bodies, see Eq. (4) in Pravec et al. (2006). The assumption that the two components have same albedo was checked by Pravec et al. (2006) for systems with best data and was found to be held to within 20%.

Since the period of recurrence of the mutual events includes a synodic effect due to a motion of the system with respect to Earth and Sun, the period is synodic. For a given pole and a radius of the mutual orbit (assumed or sampled on some grid), however, sidereal orbital period $P_{\mathrm{orb}}^{\mathrm{sid}}$ can be derived from instants of time when certain mutual events begin or end. The instants are called *contacts*[4] and a notation used is following: the first and the fourth contacts denote the beginning and the end of the partial event, respectively, while the second and the third contacts denote the beginning and the end of the total event, respectively.

First, the contacts have to be identified in the long-period component of the lightcurve. This is done visually on a basis of rapid decrease or increase of the brightness at the beginning or the end of the event, respectively; cases where the change is loosely defined[5] are omitted in the analysis.

All identified contacts in the lightcurve are sorted by the time of their occurrence, T_i ($i = 1, \ldots, N$, where N is the number of contacts), and for the given components size ratio, mutual orbit pole and size, unit vectors $\vec{c}^0$ pointing to a secondary's center in the contacts are computed (see Appendix B).[6] Arguments of mean lengths, L, corresponding to $\vec{c}^0$ are then computed as oriented angles between the ascending node and $\vec{c}^0$.[7]

The arguments are then converted to time-increasing (not restricted to interval $[0, 2\pi)$ only) values defined as

$$L' = L + k2\pi, \tag{2}$$

where k is the number of cycles[8] that passed from T_1.

The sidereal orbital period $P_{\mathrm{orb}}^{\mathrm{sid}}$ of the secondary and L_0, argument of mean length for $T = 0$, are then expressed by linear fitting of pairs of $[L_i', T_i]$, that obey a following form

$$L' = nT + L_0, \tag{3}$$

where $n = 2\pi / P_{\mathrm{orb}}^{\mathrm{sid}}$ is an angular velocity, or mean motion, of the secondary.

The above-described algorithm can be applied for all values of mutual orbit pole and radius sampled on some grid.

Since it is based on fitting of just a few time instants identified in the lightcurve, it allows a construction of a map of initial values of the parameters to be time-effective.

2.1. Lightcurve computation—the direct problem

We represent shapes of both components with spheres, ellipsoids, or other arbitrary shapes approximated by polyhedra with triangular facets. Their vertices are distributed so that the facets have similar areas in the case of spherical shape.

In modeling the long-period component of the lightcurve, a rotationally symmetric shape is used for the primary in order to flatten out its rotational component. The shape of the primary is assumed to be an oblate spheroid, with a spin axis assumed to be normal to the orbital plane of the secondary. Detailed modeling of radar data obtained for binary system 1999 KW$_4$ (see Ostro et al., 2006) showed that the primary's rotational pole and the mutual orbit normal have same direction within their error bars,[9] confirming the assumption for the case. A flattening of the primary shape is expressed as its equatorial-to-polar semiaxes ratio, A_1/C_1.

In order to simulate an amplitude of the rotational part of the long-period component of the lightcurve, a shape of the secondary is taken as a prolate spheroid synchronously rotating so that its long axis is aligned with the centers of the two bodies at the pericenter. An elongation of the secondary shape is expressed as

[1] The components are independent and additive if an effect of mutual illumination between the two bodies is neglected.

[2] F_{mut}, if expressed in intensities, is always negative.

[3] Similarly, the terms "primary event" and "secondary event" are used according to which body is being occulted/eclipsed.

[4] These contacts are apparent only and do not have any relation to "contact binaries."

[5] This occurs especially for the second and the third contacts of primary events due to the possible mutual overlap of occultations and eclipses.

[6] Unit vectors are marked with the ^{0} exponent throughout this work.

[7] The argument of mean length is defined as $L = \omega + M$, where ω is an argument of pericenter and M is a mean anomaly.

[8] This number is calculated using the first guess of orbital period of $P_{\mathrm{orb}}^{\mathrm{syn}}$.

[9] Standard errors for primary's rotational pole and mutual orbit pole was estimated to be 3 and ∼1.5 degrees, respectively.

its equatorial (the largest) to polar (the shortest) semiaxes ratio, A_2/C_2.

A size ratio of the components is expressed as a ratio of their largest semiaxes, A_2/A_1.

The system is dynamically treated as two point masses orbiting each other without external forces, i.e., the orbit is Keplerian and its pole is fixed in the inertial reference frame.[10]

The orientation and the shape of mutual orbit are described by five parameters: pole coordinates in ecliptic frame λ_p, β_p (Epoch J2000), the semimajor axis-to-primary equatorial semiaxis ratio a/A_1, the eccentricity e, and the argument of pericenter ω. Since a bulk density of the bodies is not known, a/A_1 and the sidereal period $P_{\mathrm{orb}}^{\mathrm{sid}}$ have to be searched as independent variables.

The advantage of using a/A_1 (instead of, e.g., a/C_1) is that for occultation/eclipse events occurring near the primary's equator, as is the case for most of observed binaries, a length of events is directly related to this parameter.

Instead of using a mean anomaly of the secondary for a given epoch T_0, we use the argument of mean length L_0 (i.e., the angular distance from the ascending node). The reason for doing so is that this parameter is not correlated with ω and could be precomputed from the contacts of observed events for given pole and semimajor axis. Moreover, in an eccentric orbit assumption, the value of L_0 derived for circular orbit can be used as the first approximation, without the need to know a location of orbit's pericenter.

The total brightness of the system as seen by the observer is defined as

$$I = D(\Delta, R, \alpha)\left(\int\limits_{\Sigma_1} J(\mu, \mu_0, \alpha) O \varpi \,\mathrm{d}s_1 + \int\limits_{\Sigma_2} J(\mu, \mu_0, \alpha) O \varpi \,\mathrm{d}s_2\right), \tag{4}$$

where Σ_1 and Σ_2 are surfaces of the primary and the secondary, respectively, J is a scattering law, μ and μ_0 are cosines of angles of emergence and incidence on the surfaces of the bodies, respectively, α is a phase angle, ϖ is an albedo and $\mathrm{d}s_1$ and $\mathrm{d}s_2$ are elements of the surfaces of the two bodies. Function O is equal to 1 or 0, depending on whether the surface element is both visible and illuminated, or not. Function D contains geo- and heliocentric distance and phase angle factors. It is omitted in our analysis since the observed data have been reduced to unit distances and to a consistent phase angle.

While the surfaces are described with the finite number of facets, the above formula can be rewritten as

$$I = \sum_{i=1}^{N_1} A_{1,i} o_{1,i} r_{1,i} \cos e_{1,i} + \sum_{i=1}^{N_2} A_{2,i} o_{2,i} r_{2,i} \cos e_{2,i}. \tag{5}$$

The expression on the right side is a sum of individual contributions from N_1 and N_2 facets of the primary and secondary, respectively. $A_{1,i}$ and $A_{2,i}$ are areas of the facets, $r_{1,i}, r_{2,i}$ are bidirectional reflectances defined by scattering law, and $e_{1,i}, e_{2,i}$ are angles of emergence of the facets. Factors $o_{1,i}, o_{2,i}$ represent a visibility, which is 1 or zero, depending on whether the facet is both visible and illuminated, or not, due to its orientation.

The occultations or eclipses by the other body are examined using ray-tracing code that checks which facets are occulted by or in shadow from the occulting/eclipsing body.

Three scattering laws are generally used for modeling of lightcurves of asteroids: Lommel–Seeliger (LS) law (see, e.g., Kaasalainen and Torppa, 2001), and two Hapke's laws for macroscopically smooth and rough surface (Bowell et al., 1989). We chose to use the Lommel–Seeliger law (which is $J = \mu_0/(\mu_0 + \mu)$) for its simplicity. In some cases, the Hapke's law for macroscopically rough surface was also tested and we found that parameters of obtained best-fit solution are only weakly sensitive to the photometric law used (see Section 3).

2.2. Interpolation of synthetic lightcurve

Since the overall shape of synthetic lightcurve is quite smooth at almost all points, it would be inefficient to compute the total brightness of binary at epochs of all observed data points, especially if the data are dense. Instead, it is more effective to divide timespans of observations into an equidistant grid, to evaluate the brightness only at this grid, and to interpolate the values at the points within.

However, at the time instants of events' contacts the condition of lightcurve smoothness is not fulfilled, and thus their epochs have to be included in the grid. The epochs are computed from true anomalies of the radius vectors of the secondary, solved for spherical approximation of the bodies' shapes using algorithm described in Appendix B.

We point out that, because the interpolation is applied to the specific synthetic lightcurve and not to the real data, all parameters needed for this evaluation (i.e., location of pericenter, eccentricity, orbital pole, etc.) are given in this case.

2.3. Deriving binary parameters—the inverse problem

The inverse problem entails the derivation of binary system parameters from the observed lightcurve data. More specifically, it is equivalent to finding all plausible minima of a function (root-mean-square)

$$\mathrm{RMS}(\mathbf{B}, \Delta m_1, \ldots, \Delta m_P) = \sqrt{\frac{\sum_{i=1}^{N}(o_i - c_i)^2}{N}}, \tag{6}$$

where $\mathbf{B}$ is vector of fitted parameters, $\Delta m_1, \ldots, \Delta m_P$ are offsets (in magnitudes) of lightcurve sequences that are not absolutely calibrated (if all the sequences are calibrated, then $P = 1$), while o_i and c_i are observed and calculated magnitudes, respectively, of N data points in the lightcurve. The term *lightcurve sequence* refers in this work to a subset of photometric measurements with common set of comparison stars within the lightcurve.

Usually, many authors use the χ^2 statistics for minimization and errors estimation. This approach assumes that the observational errors are random and that the model gives a good description of the data, but neither is guaranteed here. The observed data themselves may contain systematic errors due to the interfering stars and changes in observational conditions and, more importantly, the model is only an approximation and thus produces systematic deviations between synthetic and observed data. Therefore, an arbitrary function of $(\sum_{i=1}^{N}(o_i - c_i)^2)$ or $(\sum_{i=1}^{N}|o_i - c_i|)$ can be used for minimization. We choose the RMS function for its explicit meaning.

Since the primaries of studied binaries are approximated with rotationally symmetric shapes, only the long-period component of their lightcurves are modeled in this work. These components are hereafter referred to as *lightcurves* for brevity.

In the simplest model with spherical bodies orbiting each other on a circular orbit, the list of fitted parameters $\mathbf{B}$ consists of five parameters: ratio of component diameters D_2/D_1, the mutual orbit semimajor axis-to-primary diameter ratio a/D_1 and its pole in ecliptic coordinates λ_p, β_p, the sidereal period $P_{\mathrm{orb}}^{\mathrm{sid}}$, and the secondary's argument of mean length L_0 at a given epoch T_0.

With increasing complexity of the model, the number of fitted parameters also increases. They could include the eccentricity and the argument of pericenter of the mutual orbit, parameters

[10] A correction for an oblateness of the primary is taken into account in determining a bulk density (Section 2.4).

Table 1
Observational circumstances of modeled binary NEAs.

	Object	Apparition	Time span (d)	Geo. arc (°)	Hel. arc (°)	Phase angle (°)
(175706)	1996 FG_3	1998 Dec–99 Jan	36	30	27	14–32
(65803)	Didymos	2003 Nov–Dec	29	38	29	2–19
(66391)	1999 KW_4	2000 May–Jun[a]	35	90	17	62–76
		2001 May–Jun	18	99	9	33–66
(185851)	2000 DP_{107}	2000 Sep–Oct	8	19	7	30–38
(66063)	1998 RO_1	2002 Sep[a]	3	6	1	3–8
		2003 Sep	9	33	5	12–33
		2004 Sep	11	28	6	30–36

Note. The fourth and the fifth columns give the sky arc spanned with respect to the Earth and to the Sun, respectively. Time span, phase angle and arcs cover only sessions at which at least part of a mutual event was observed.
[a] Denotes apparitions with data of low quality or too short time span, that were only checked for consistency with solutions derived from other apparition(s).

describing the shape, rotational periods and spin poles of both components, etc.

The minimization process is divided into several steps:

I. First, an initial value (or its lower limit) of ratio of effective diameters at equatorial aspect, D_2/D_1 is derived from the lightcurve using Eq. (4) from Pravec et al. (2006), the contacts of events are identified visually in the lightcurve, and the synodic orbital period P_{orb}^{syn} is derived from the periodicity of the events.
II. An equidistant grid in β_p and $\lambda' = \lambda_p \cos\beta_p$[11] is constructed in the next step. For each point of the grid, a reasonable interval of the orbit semimajor axis a/D_1 is sampled and for each combination of β_p, λ_p and a/D_1 the sidereal period P_{orb}^{sid} with the argument of mean length L_0 of the secondary are computed. The value of a/D_1 with the best fit of Eq. (3) to the epochs of contacts identified in the lightcurve is then selected as the most probable value. If some observed contact could not occur for given pair of λ_p, β_p and any of the tested values of a/D_1, the pair is excluded from further analysis.
III. The above-presented process serves to create a map of initial values of parameters D_2/D_1, λ_p, β_p, a/D_1, P_{orb}^{sid} and L_0. Starting from each point of this map, a local minimum of the RMS function is found using the Nelder and Mead Simplex Algorithm (Buchanan and Turner, 1992) for spherical components and circular orbit approximation.
IV. In order to make subsequent computation faster, only a subset of found local minima is selected for further analysis. The minima are selected from those with the lowest RMS to those with the largest, while a distance in λ_p, β_p space of each selected minimum from all previous has to be at least 10 degrees.
V. For each set of parameters of these selected minima a grid in e and ω is created. This grid serves as initial points for the next minimization using the simplex algorithm, where fitted parameters are $\mathbf{B} = (D_2/D_1, \lambda_p, \beta_p, a/D_1, P_{orb}^{sid}, L_0, e, \omega)$.
VI. The same selection process as in the step IV is then applied to all found local minima.
VII. A grid in A_1/C_1 and A_2/C_2 is then created for all selected minima and a minimization is restarted from this grid in order to derive the flattening of the primary and the elongation of the secondary. Fitted parameters are $\mathbf{B} = \lambda_p, \beta_p, a/A_1, P_{orb}^{sid}, L_0, e, \omega, A_1/C_1, A_2/C_2, A_2/A_1$. An initial values of the component size ratio and semimajor axis are set as $A_2/A_1 = D_2/D_1$ and $a/A_1 = 2(a/D_1)$, respectively.
VIII. The above described process generally ends up with a number (up to a few tens) of local minima. Some of them have significantly higher rms residuals and poorly fitting synthetic lightcurves, so they are just spurious minima and not real solutions. Remaining local minima typically cluster in small volumes of the parameters space and they appear to be just fluctuations in the rms function due to limited number of

[11] This parametrization ensures that an equal area is belonging to each point of the λ_p, β_p grid.

Table 2
Estimated parameters of binary NEAs, with 3-σ errors.

Object (apparition)	Solution	RMS	D_2/D_1	a/A_1	λ_p (°)	β_p (°)	P_{orb}^{sid} (h)	e
(175706) 1996 FG_3 (1998)	I	0.018	$0.28^{+0.01}_{-0.02}$	$3.1^{+0.9}_{-0.5}$	242^{+96}_{-96}	-84^{+14}_{-5}	$16.14^{+0.01}_{-0.01}$	$0.10^{+0.12}_{-0.10}$
(65803) Didymos (2003)	I	0.012	$0.22^{+0.01}_{-0.01}$	$2.8^{+0.2}_{-0.2}$	157^{+4}_{-7}	$+19^{+45}_{-15}$	$11.906^{+0.004}_{-0.01}$	$0.09^{+0.07}_{-0.09}$
	II	0.012	$0.21^{+0.01}_{-0.01}$	$2.9^{+0.3}_{-0.2}$	329^{+11}_{-194}	-70^{+25}_{-15}	$11.920^{+0.004}_{-0.006}$	$0.02^{+0.01}_{-0.02}$
(66391) 1999 KW_4 (2001)	I	0.036	$0.46^{+0.06}_{-0.06}$	$3.2^{+0.6}_{-0.5}$	341^{+9}_{-7}	-56^{+20}_{-18}	$17.42^{+0.01}_{-0.03}$	$0.04^{+0.09}_{-0.04}$
(185851) 2000 DP_{107} (2000)	I	0.027	$0.43^{+0.3}_{-0.04}$	$5.0^{+2.0}_{-0.3}$	291^{+18}_{-26}	$+80^{+7}_{-40}$	$42.09^{+0.13}_{-0.08}$	$0.05^{+0.19}_{-0.05}$
	II	0.026	$0.43^{+0.2}_{-0.05}$	$4.9^{+1.2}_{-0.2}$	31^{+23}_{-12}	-61^{+18}_{-10}	$42.79^{+0.09}_{-0.11}$	$0.09^{+0.06}_{-0.06}$
(66063) 1998 RO_1 (2003)	I	0.027	$0.48^{+0.03}_{-0.03}$	$3.0^{+0.7}_{-0.4}$	277^{+180}_{-180}	$+37^{+53}_{-2}$	$14.54^{+0.02}_{-0.01}$	$0.04^{+0.08}_{-0.04}$
(66063) 1998 RO_1 (2004)	I	0.025	$0.48^{+0.03}_{-0.03}$	$3.5^{+0.7}_{-0.6}$	274^{+17}_{-73}	48^{+28}_{-6}	$14.54^{+0.03}_{-0.02}$	$0.06^{+0.04}_{-0.06}$

Object (apparition)	Solution	ω (°)	L_0 (°)	Epoch (MJD)	A_2/A_1	A_1/C_1	A_2/C_2	ρ ($g\,cm^{-3}$)
(175706) 1996 FG_3 (1998)	I	79^{+70}_{-93}	43^{+96}_{-96}	51150.6792	$0.33^{+0.07}_{-0.08}$	$1.2^{+0.5}_{-0.2}$	$1.4^{+0.3}_{-0.2}$	$1.4^{+1.5}_{-0.6}$
(65803) Didymos (2003)	I	349^{+13}_{-34}	120^{+3}_{-6}	52963.8922	$0.21^{+0.02}_{-0.02}$	$1.0^{+0.3}_{-0.0}$	$1.0^{+0.1}_{-0.0}$	$1.7^{+0.6}_{-0.4}$
	II	174^{+95}_{-25}	301^{+10}_{-191}	”	$0.20^{+0.02}_{-0.01}$	$1.1^{+0.2}_{-0.1}$	$1.0^{+0.1}_{-0.0}$	$2.1^{+0.8}_{-0.5}$
(66391) 1999 KW_4 (2001)	I	264^{+67}_{-64}	129^{+9}_{-8}	52054.0398	$0.51^{+0.08}_{-0.16}$	$1.1^{+0.9}_{-0.1}$	$1.2^{+0.2}_{-0.2}$	$1.2^{+1.1}_{-0.5}$
(185851) 2000 DP_{107} (2000)	I	328^{+16}_{-11}	270^{+20}_{-11}	51812.1522	$0.44^{+0.16}_{-0.09}$	$1.2^{+1.1}_{-0.2}$	$1.2^{+0.1}_{-0.1}$	$0.8^{+1.1}_{-0.1}$
	II	161^{+23}_{-19}	89^{+26}_{-18}	”	$0.37^{+0.20}_{-0.10}$	$1.6^{+1.6}_{-0.6}$	$1.1^{+0.2}_{-0.1}$	$1.1^{+2.5}_{-0.6}$
(66063) 1998 RO_1 (2003)	I	182^{+87}_{-176}	252^{+179}_{-180}	52898.8447	$0.56^{+0.09}_{-0.09}$	$1.2^{+0.6}_{-0.2}$	$1.4^{+0.3}_{-0.1}$	$1.5^{+1.7}_{-0.6}$
(66063) 1998 RO_1 (2004)	I	188^{+68}_{-15}	201^{+74}_{-12}	53258.1605	$0.45^{+0.25}_{-0.04}$	$2.1^{+0.4}_{-1.1}$	$1.5^{+0.6}_{-0.1}$	$4.1^{+0.8}_{-2.8}$

Note. The parameters are described in Section 2. RMS is a root mean square of residuals between observed and synthetic data, in magnitudes. L_0 is expressed for a given epoch.

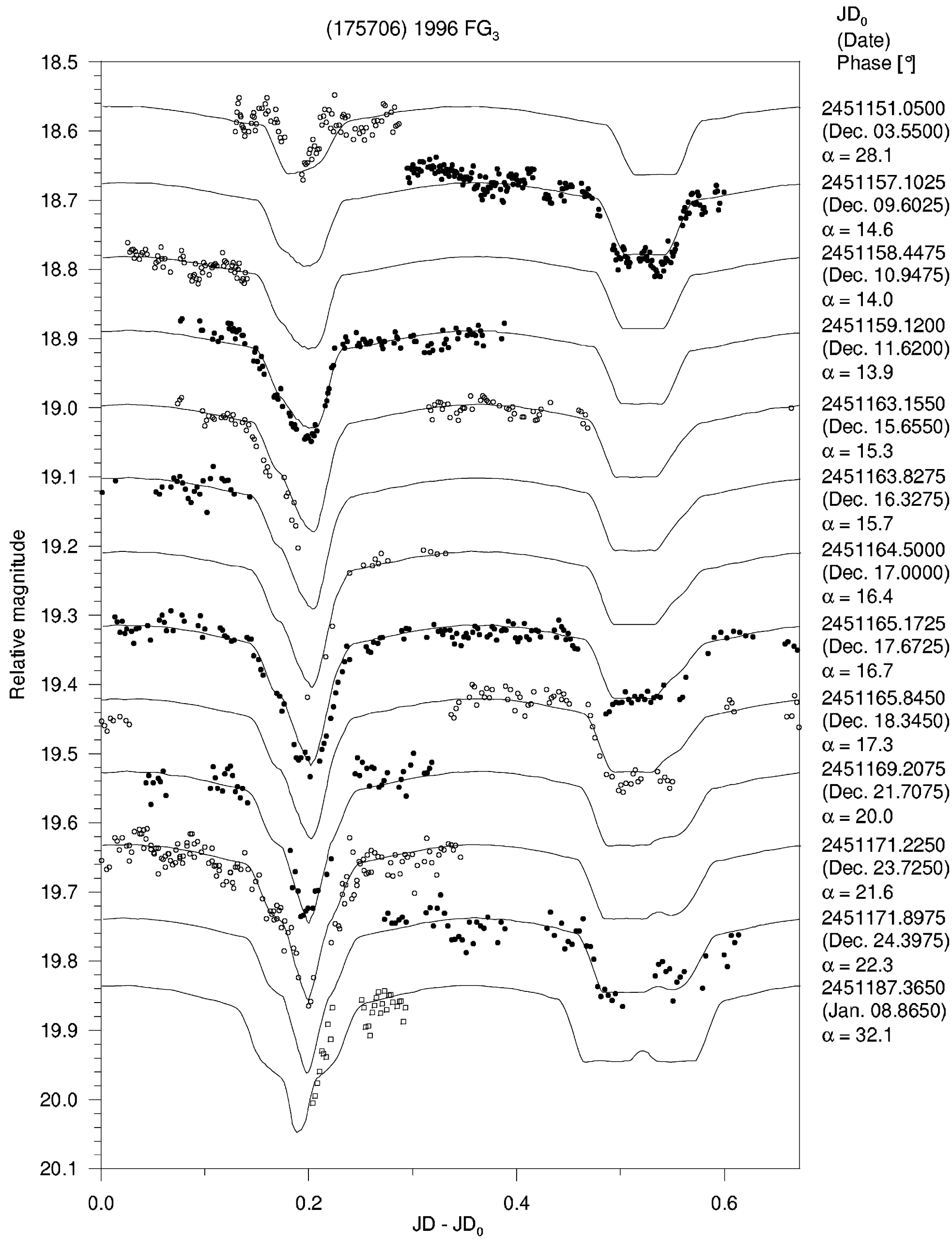

Fig. 1. Long-period lightcurve component of (175706) 1996 FG$_3$. The observational data (points) are plotted together with the synthetic lightcurve for the best-fit solution (curve). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.1 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD$_0$) are listed in the right column for each curve, as well as the phase angle α.

Table 3
Type of events occurring in the lightcurves for the solutions from Table 2.

Object (apparition)	Solution	Events type
(175706) 1996 FG$_3$ (1998)	I	Both
(65803) Didymos (2003)	I	Both
	II	Both
(66391) 1999 KW$_4$ (2001)	I	Eclipses (except the event at JD = 2452056.45)
(185851) 2000 DP$_{107}$ (2000)	I	Occultations
	II	Eclipses
(66063) 1998 RO$_1$ (2003)	I	Both
	II	Both
(66063) 1998 RO$_1$ (2004)	I	Both

degrees of freedom. We choose the best one as the nominal solution; the other minima fall into a plausibility range around the nominal one (see below).

2.4. Bulk density

Kepler's third law can be written in the form

$$\frac{4\pi^2}{G P_{\rm orb}^{\rm sid\,2}}\left[\frac{M_2}{M_1}+1\right]^{-1}\frac{a^3}{V_1}=\rho, \qquad (7)$$

where ρ is the bulk density of the primary, G is the gravitational constant, M_2 and M_1 are masses of the secondary and the primary, respectively, and V_1 is a volume of the primary.

Assuming that both components have same bulk density, the term M_2/M_1 can be replaced with a ratio of volumes of secondary and primary, V_2/V_1. This gives[12]

[12] For spherical bodies, $\frac{24\pi}{G P_{\rm orb}^{\rm sid\,2}}[(D_2/D_1)^3+1]^{-1}(a/D_1)^3=\rho$.

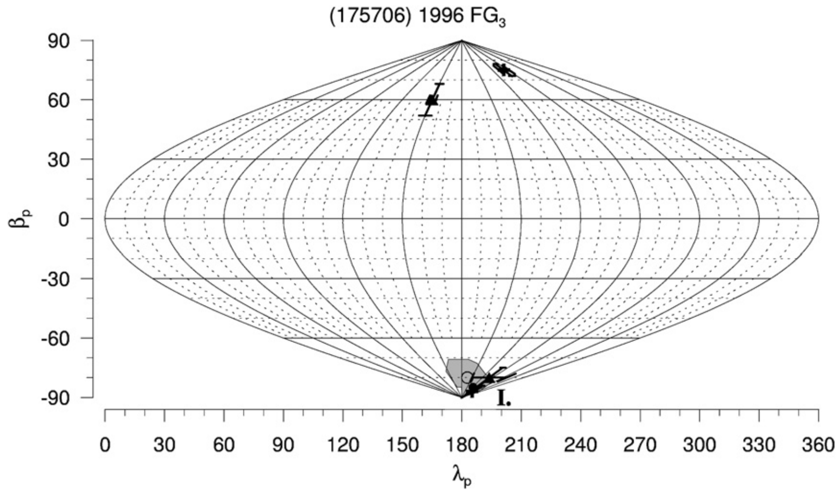

Fig. 2. Range of plausible poles of the mutual orbit of (175706) 1996 FG_3 in ecliptic coordinates. The best-fit solutions from this work (filled circle), from Mottola and Lahulla (2000) (crosses), from Pravec et al. (2000) (open circle) and from Scheirich (2003) (filled triangles) are denoted as well. 1-σ error bars are indicated for Mottola and Lahulla (2000) and Scheirich (2003) solutions.

$$\frac{4\pi^2}{G {P_{\rm orb}^{\rm sid}}^2}\left[\frac{V_2}{V_1}+1\right]^{-1}\frac{a^3}{V_1}=\rho. \tag{8}$$

Using the above formula without caution for spherical approximation of bodies is the largest source of error in derived densities and it often leads to unrealistically low values.

If the primary shape is assumed to be homogeneous oblate ellipsoid and considering the effect of the oblateness up to the J_2 term (the first zonal harmonic coefficient in the gravitational potential expansion), Kepler's third law can be modified as follows (see Rossi et al., 1999 or Murray and Dermott, 1999, p. 268):

$$\frac{4\pi^2 a^3}{G {P_{\rm orb}^{\rm sid}}^2 (V_1+V_2)}\epsilon=\rho, \tag{9}$$

where

$$\epsilon = 1-\frac{3(I_C-I_A)}{2a^2M_1}=1-\frac{3}{10}\frac{A_1^2-C_1^2}{a^2}, \tag{10}$$

with I_C and I_A the moments of inertia of oblate ellipsoid with respect to the C and A axes, respectively.

Photometric observations usually do not allow to derive the volumes of the bodies accurately, and thus their values (especially V_1) in Eq. (9) is a significant source of uncertainty in estimating the bulk density.

2.5. Assessment of parameters' uncertainties

A special attention is given to assessing uncertainties of estimated parameters. Due to the model simplifications and systematic errors, formal χ^2-error estimates give unrealistically small uncertainties. Thus, all fitted parameters were stepped in a reasonable interval around their best-fit solution values and held fixed, while remaining parameters were fitted by the simplex algorithm. Synthetic lightcurves were constructed for each result of this fitting and visually compared with the data. Parameters respective to the plausible selected curves represent 3-σ ranges of the solutions.

In addition, for easier comparison of orbital poles obtained for different apparitions and with radar-derived poles, λ_p and β_p were stepped in a two-dimensional grid and ranges of possible values are shown in plots in Section 3.

The uncertainty of the bulk density can not be estimated using the above method, because the density is not among directly fitted parameters, but it is derived from them. A range of possible densities can be obtained by varying these parameters separately, but in order not to miss their combination leading to extremal values of density, a following method has been used:

An auxiliary parameter ρ_0 is stepped in a reasonable interval, usually [0.5, 4.0]. For every value of ρ_0, the parameters describing the binary system are varied in order to reach a minimum of a function

$$X = \mathrm{RMS}\cdot\left(1+w\left|\rho(\mathbf{B})-\rho_0\right|\right), \tag{11}$$

where RMS is defined by Eq. (6) and ρ is considered as a function of varied parameters defined by Eq. (9). Several values of the weighting factor w in a range from 0.5 to 2.0 were used. The purpose of this procedure is to obtain a parameter set not just with the value of ρ close or equal to stepped parameter ρ_0, but also with the best possible agreement to the observed data. After each fitting, synthetic curve was constructed using the resulting parameters, compared visually with the observed data, and the 3-σ range of ρ was extended in a case of a good match.

3. Modeled systems

We modeled five near-Earth binary asteroids, three of them in multiple apparitions. The binaries are listed in Table 1 with their observational circumstances. The best-fit values of modeled parameters and their errors for five best cases are summarized in Table 2 and the results for these systems are summarized in this section in more detail. Table 3 lists types of events occurring in their lightcurves, determined from estimations of their mutual orbits' normals.

Although solutions derived from radar data or from optical data obtained during other apparitions exist for some cases, the presented results are based solely on photometric data taken during given apparitions. This helps to get better understanding of effects affecting an accuracy of derived parameters, mainly of the position of the mutual orbit's pole, which will be crucial when larger sample of binaries data will be available and a debiasing of their orbital pole distribution will be possible. Results from other works are cited as well.

All dates and times used in this section are in the UTC and have been corrected for light-travel time.

3.1. (175706) 1996 FG_3

The binary nature of this near-Earth asteroid was discovered by Pravec et al. (2000) and by Mottola and Lahulla (2000). The

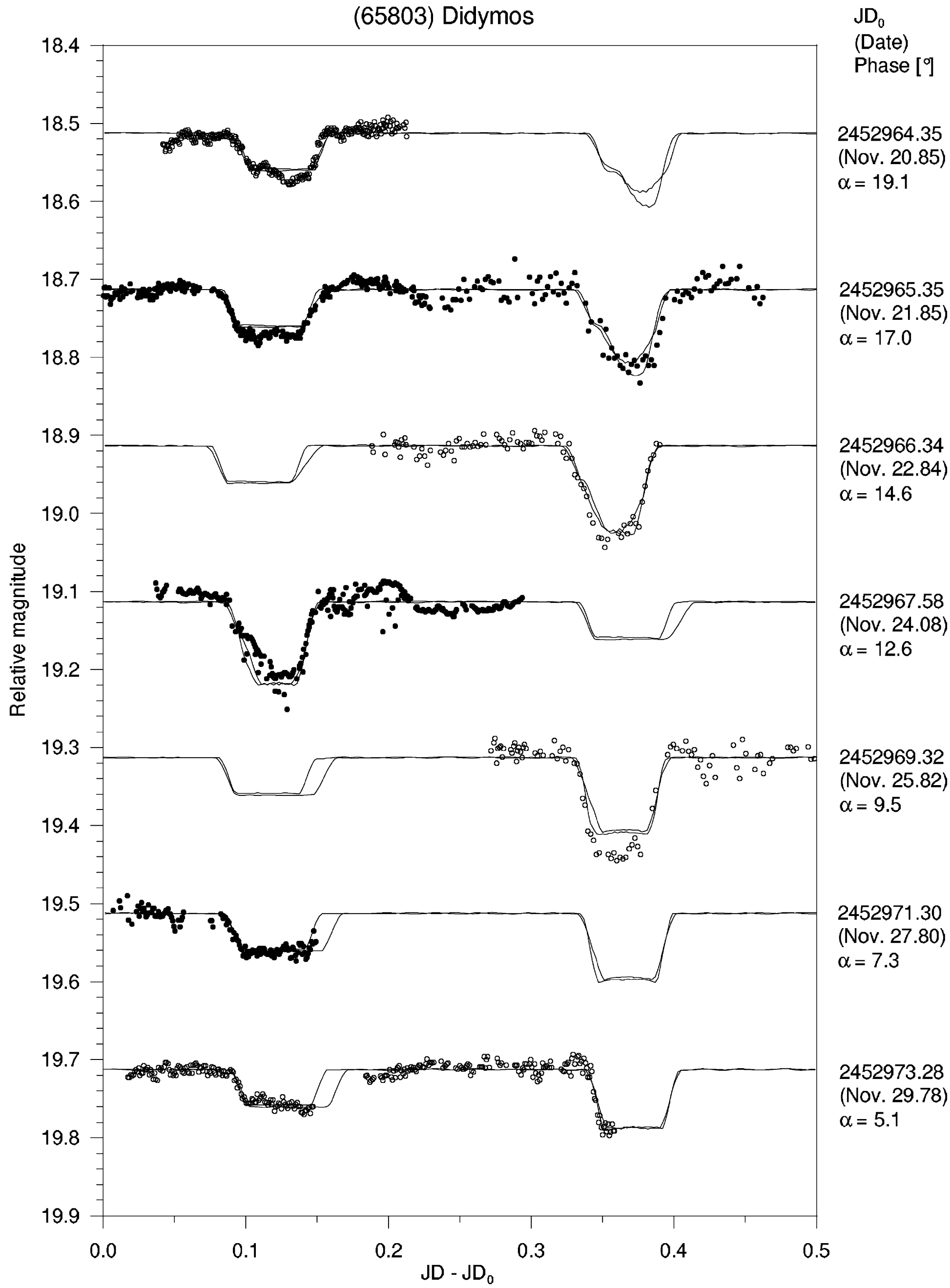

Fig. 3. Long-period lightcurve component (part I) of (65803) Didymos. The observational data (points) are plotted together with the synthetic lightcurves for the best-fit solutions (curves). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.2 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD_0) are listed in the right column for each curve, as well as the phase angle α. On the fourth curve from the top, the minima are shown in an order opposite to other curves.

results of both works were in agreement and they are summarized here. Pravec et al. (2000) derived $D_2/D_1 = 0.31 \pm 0.02$, $a/D_1 = 1.7 \pm 0.3$, $P_{\mathrm{orb}}^{\mathrm{sid}} = (16.135 \pm 0.005)$ h and $e = 0.05 \pm 0.05$ from their lightcurve. Assuming that $D_2/D_1 = 0.3$, $a/D_1 = 1.55$, $P_{\mathrm{orb}}^{\mathrm{sid}} = 16.135$ h, $\lambda_{\mathrm{p}} = 196°$ and $\beta_{\mathrm{p}} = -80°$ (they did not carry out the complete inverse search of parameters) they constructed a simple model with spherical bodies and a circular mutual orbit with synthetic lightcurve with a good match to observational data.

Mottola and Lahulla (2000) developed a numerical model consisting of two triaxial ellipsoids on a circular orbit. They determined $P_{\mathrm{orb}}^{\mathrm{sid}}$ to be 16.15 ± 0.02 h and obtained two solutions for orbital pole: ($\lambda_{\mathrm{p}} = 282° \pm 10°$, $\beta_{\mathrm{p}} = -87° \pm 3°$) and ($\lambda_{\mathrm{p}} = 262° \pm 10°$, $\beta_{\mathrm{p}} = +75° \pm 3°$), with the retrograde solution being more likely. The component shapes properties and size of the orbit were the same for both solutions. Since Mottola and Lahulla used a slightly different parametrization from that used in this paper, we converted their values to the system defined in Section 2 for easier comparison, approximating the equatorial radius of primary as $A_1 = \sqrt{AB}$, where A, B are equatorial semiaxes of primary defined by Mottola and Lahulla. These converted parameters are as follows, and are in a good agreement with our results (see Table 2):

$$A_1/C_1 = 1.4 \pm 0.2,$$

$$A_2/C_2 = 1.4 \pm 0.2,$$

$$A_2/A_1 = 0.32 \pm 0.03,$$

$$a/A_1 = 2.9 \pm 0.1.$$

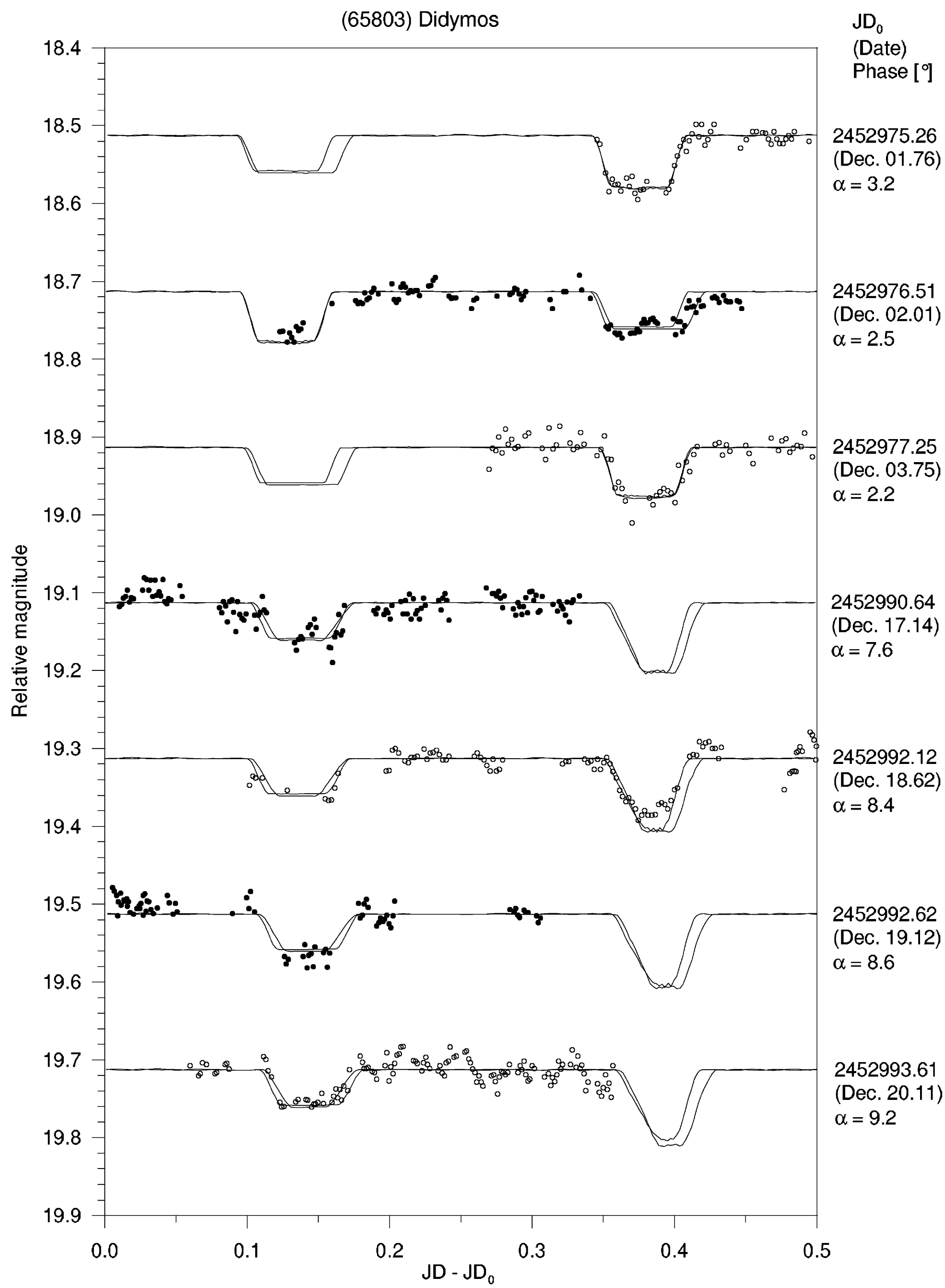

Fig. 4. Long-period lightcurve component (part II) of (65803) Didymos. The observational data (points) are plotted together with the synthetic lightcurves for the best-fit solutions (curves). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.2 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD$_0$) are listed in the right column for each curve, as well as the phase angle α. On the second curve from the top, the minima are shown in an order opposite to other curves.

Scheirich (2003) found two solutions from the Ondřejov data using spherical bodies on eccentric orbit: ($\lambda_p = 260° \pm 50°$, $\beta_p = -80° \pm 5°$) and ($\lambda_p = 150° \pm 4°$, $\beta_p = +60° \pm 8°$).

We reanalysed the data by Pravec et al. (2000) taken during 1998-12-03.7 to 1999-01-09.1 using the model of two-axis ellipsoids on the eccentric orbit described in Section 2.1 and found unique solution fitting the long-period component of the lightcurve satisfactorily. The parameters of this solution are summarized in Table 2, with their errors obtained using procedure described in Section 2.5. The Lommel–Seeliger (LS) photometric law was used in the modeling. Data from Dec. 9/10 and from Dec. 18.68 to 19.03 were taken as relative in order to describe depths of secondary events properly. The shifts obtained for these two sessions was −0.015 and −0.010 mag. For comparison of different scattering laws, the minimization was carried out using the Hapke's law for macroscopically rough surface also, but the obtained parameters did not differ significantly.[13]

The long-period component of the data together with the synthetic lightcurve of the best-fit solution are presented in Fig. 1. The region of plausible values of λ_p and β_p is plotted in Fig. 2 together with the best-fit solutions of all cited works.

[13] We used parameters of Hapke's law which Bowell et al. (1989) derived for C-class asteroid, 24 Themis. See their Table IV.

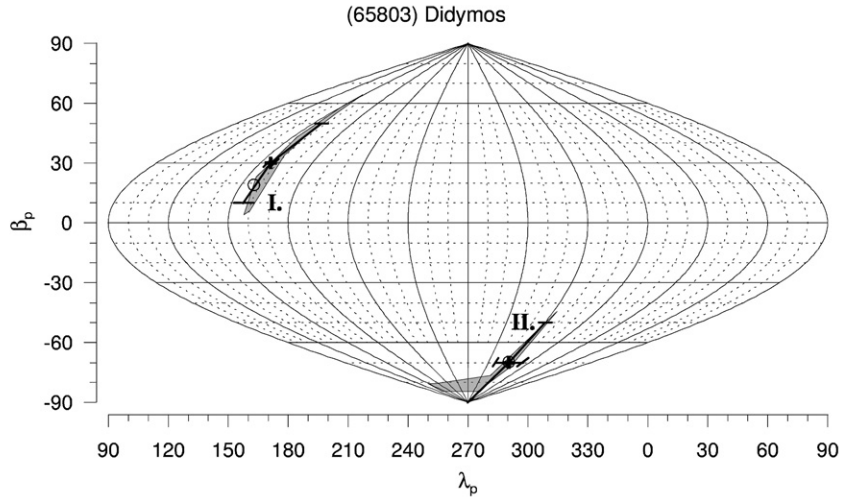

Fig. 5. Range of plausible poles of the mutual orbit of (65803) Didymos in ecliptic coordinates. The best-fit solutions from this work (open circles) and from Pravec et al. (2006) (crosses, with 3-σ formal error bars indicated) are denoted as well.

3.2. (65803) Didymos

The binary nature of this near-Earth asteroid was discovered by Pravec et al. (2003b) and its lightcurve data were investigated by Pravec et al. (2006) in detail. They constructed a numerical model consisting of two spherical bodies on a circular orbit and obtained two solutions. The estimated parameters of these two solutions are summarized here (λ_p, β_p in degrees): I. $P_{orb}^{sid} = 11.900 \pm 0.005$ h, $D_2/D_1 = 0.22 \pm 0.02$, $a/D_1 = 1.4 \pm 0.1$, $e = 0.06^{+0.06}_{-0.04}$, $\lambda_p = 156 \pm 2$, $\beta_p = +30 \pm 20$; II. $P_{orb}^{sid} = 11.920 \pm 0.005$ h, $D_2/D_1 = 0.21 \pm 0.02$, $a/D_1 = 1.5^{+0.2}_{-0.1}$, $e = 0.02^{+0.03}_{-0.02}$, $\lambda_p = 330 \pm 20$, $\beta_p = -70 \pm 20$.

We reanalysed the data by Pravec et al. (2006) taken during 2003-11-20.9 to 12-20.5 using the model of two-axis ellipsoids on the eccentric orbit described in Section 2.1. We found two solutions fitting the long-period component of the lightcurve satisfactorily. The parameters of these solutions, confirming the previous results, are summarized in Table 2, with their errors obtained using procedure described in Section 2.5. The LS photometric law was used in the modeling and all sessions of the data were taken as absolutely calibrated (i.e., $P = 1$ in Eq. (6)). For comparison of different scattering laws, the minimization was carried out using the Hapke's law for macroscopically rough surface also, the obtained parameters are within 3-σ errors of values derived using the LS law.[14]

The long-period component of the data together with the synthetic lightcurves of the best-fit solutions are presented in Figs. 3 and 4. The regions of plausible values of λ_p and β_p are plotted in Fig. 5 together with the best-fit solutions from this work and from Pravec et al. (2006).

3.3. (66391) 1999 KW$_4$

The binary nature of this near-Earth asteroid was discovered by Benner et al. (2001) from radar data and by Pravec and Šarounová (2001) from photometry. The radar data were investigated by Ostro et al. (2006) and Scheeres et al. (2006). Ostro et al. constructed detailed shape models of both components and derive mutual orbit parameters as follows: $P_{orb}^{sid} = 17.42 \pm 0.04$ h, $\lambda_p = 326° \pm 3°$, $\beta_p = -62° \pm 2°$, $e = 0.0004 \pm 0.0019$. Since they used general shapes of components, we converted characteristics of these shapes and the mutual orbit semimajor axis to the system defined in Section 2 for easier comparison, approximating the equatorial radius of primary as $A_1 = \sqrt{(X_1 Y_1)}/2$, polar radius of primary as $C_1 = Z_1/2$, polar radius of secondary as $C_2 = \sqrt{(Y_2 Z_2)}/2$ and the longest semiaxis of the secondary as $A_2 = X_2/2$, where $X_1, Y_1, Z_1, X_2, Y_2, Z_2$ are extents along principal axes of the shape models of the primary and the secondary, respectively, taken from their Table 2.

These converted parameters are:

$$A_1/C_1 = 1.12 \pm 0.04,$$

$$A_2/C_2 = 1.4 \pm 0.1,$$

$$A_2/A_1 = 0.38 \pm 0.02,$$

$$a/A_1 = 3.37 \pm 0.07.$$

Pravec et al. (2006) analyzed photometric data obtained in June 2000 and from May to June 2001 and derived $P_{orb}^{syn} = 17.44 \pm 0.01$ h and a lower limit on D_2/D_1 of 0.3.

We reanalysed the data by Pravec et al. (2006) taken during 2001-05-25.0 to 06-20.9 using the model of two-axis ellipsoids on the eccentric orbit described in Section 2.1 and found a unique solution, close to the solution from radar, fitting the long-period component of the lightcurve satisfactorily. The parameters of this solution are summarized in Table 2, with their errors obtained using procedure described in Section 2.5. The LS photometric law was used in the modeling and all sessions of the data were taken as absolutely calibrated (i.e., $P = 1$ in Eq. (6)). For comparison of different scattering laws, the minimization was carried out using the Hapke's law for macroscopically rough surface also, but the obtained parameters did not differ significantly.[15]

The long-period component of 2001 data together with the synthetic lightcurve of the best-fit solution are presented in Fig. 7. The region of plausible values of λ_p and β_p is plotted in Fig. 8 together with the best-fit solution from 2001 photometric data and from Ostro et al. (2006). The region for 2000 apparition (see below) is shown there as well.

Due to insufficient quality and coverage of the 2000 data (taken during 2000-05-24.9 to 06-29.0, see Pravec et al., 2006, for details), we were unable to run the full analysis of them and they were used for checking consistency with the model from the 2001 data

[14] We used parameters of Hapke's law which Bowell et al. (1989) derived for M-class asteroid, 69 Hesperia. See their Table IV.

[15] We used parameters of Hapke's law which Bowell et al. (1989) derived for S-class asteroid, 82 Alkmene. See their Table IV.

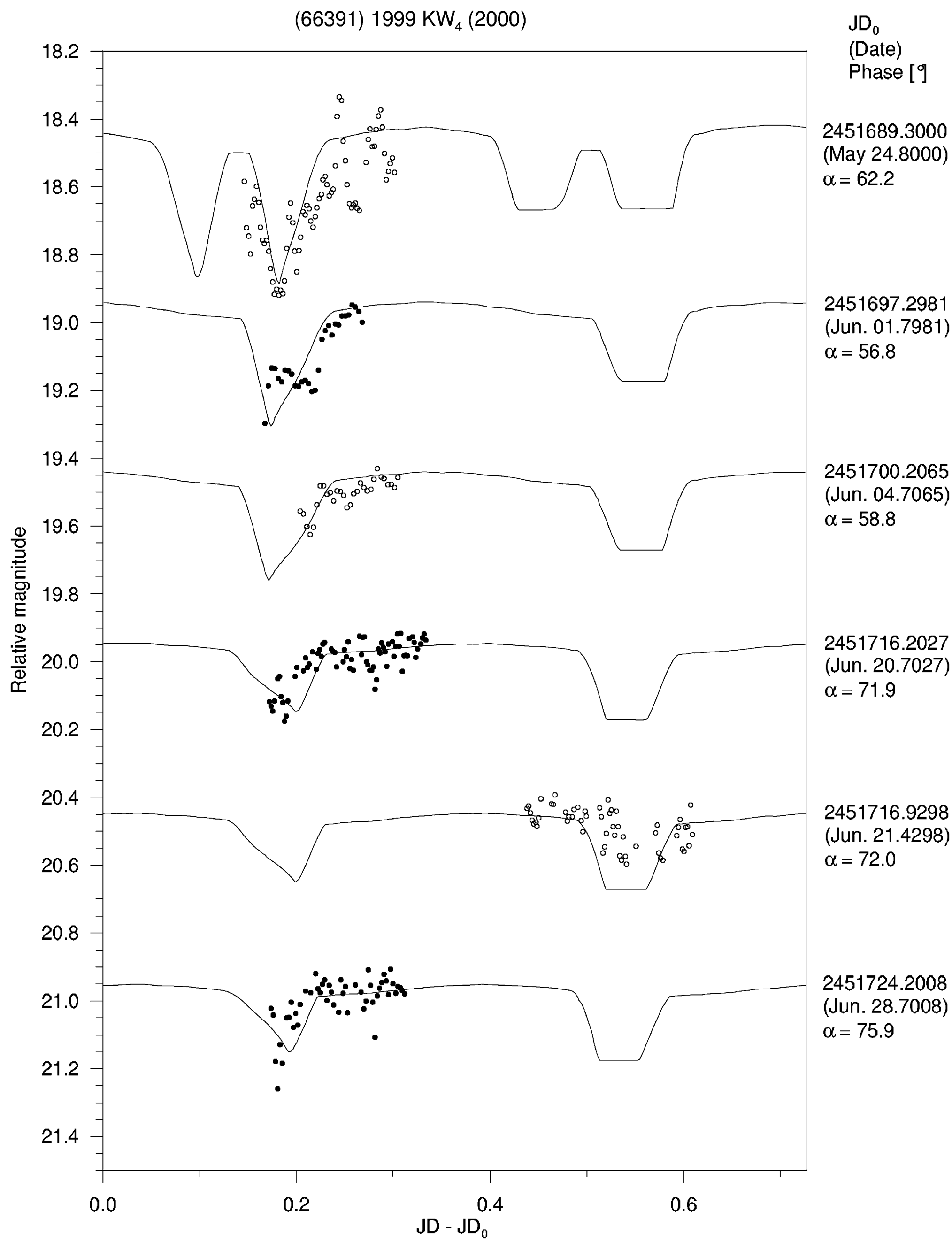

Fig. 6. Long-period lightcurve component of (66391) 1999 KW_4 (2000 apparition). The observational data (points) are plotted together with the synthetic lightcurve (curve) constructed using the pole from Ostro et al. (2006) and the semimajor axis and shape parameters from the best-fit solution of 2001 photometric data. The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.5 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD_0) are listed in the right column for each curve, as well as the phase angle α.

only. The shape parameters of components and the orbit's semimajor axis were fixed at the best-fitting values from the 2001 apparition, a circular orbit was assumed, and λ_p and β_p were stepped in a reasonable neighborhood around the 2001 best-fit solution, while P_{orb}^{sid} and L_0 were fitted. The region of plausible values of λ_p and β_p was then constructed using procedure described in Section 2.5 and is plotted in Fig. 8. The solutions obtained from photometric data from the 2000 and 2001 apparitions, and the radar solution by Ostro et al. overlap within their 3-σ error bars.

The long-period component of 2000 data together with the synthetic lightcurve constructed using the pole from Ostro et al. and the semimajor axis and shape parameters from the best-fit solution of 2001 photometric data are presented in Fig. 6.

3.4. (185851) 2000 DP_{107}

The binary nature of this near-Earth asteroid was discovered by Ostro et al. (2000) and Margot et al. (2000) from radar data and its lightcurve data were investigated by Pravec et al. (2006) in detail. Margot et al. (2002) analyzed the radar data obtained in September 2000 with the following results (λ_p, β_p in degrees): $P_{orb}^{sid} = 42.12 \pm 0.05$ h, $D_2/D_1 = 0.4 \pm 0.2$, $a/D_1 = 3.3 \pm 0.7$, $e = 0.010 \pm 0.005$, $\lambda_p = 280 \pm 27$ and $\beta_p = +73 \pm 7$.

Pravec et al. (2006) estimated synodic orbital period as $P_{orb}^{syn} = 42.2 \pm 0.1$ h and D_2/D_1 as 0.41 ± 0.02.

We reanalysed their data taken during 2000-09-25.2 to 10-05.2 using the model of two-axis ellipsoids on the eccentric orbit de-

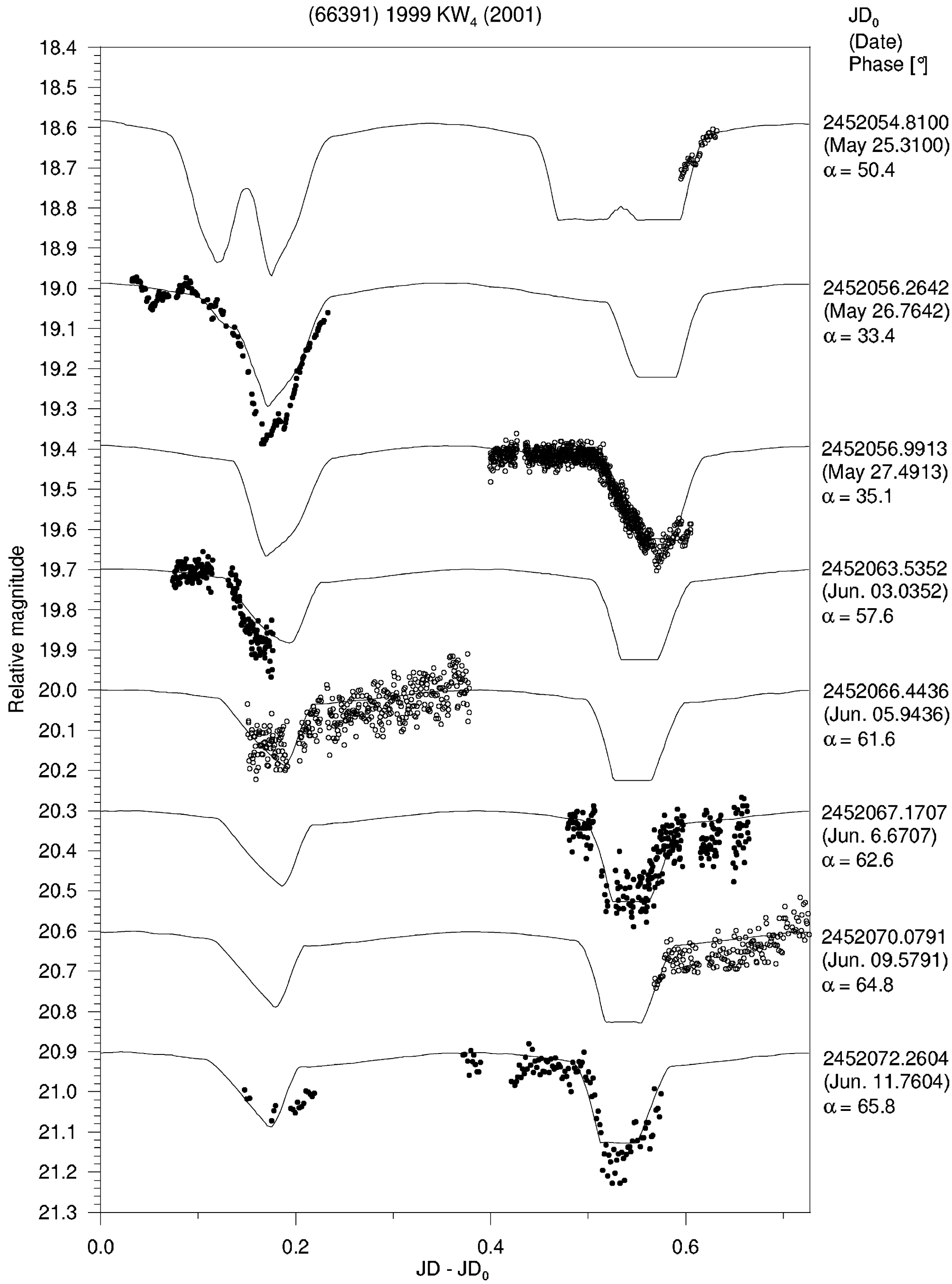

Fig. 7. Long-period lightcurve component of (66391) 1999 KW$_4$ (2001 apparition). The observational data (points) are plotted together with the synthetic lightcurve for the best-fit solution (curve). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.4 mag (first three curves from the top) and 0.3 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD$_0$) are listed in the right column for each curve, as well as the phase angle α.

scribed in Section 2.1. We found two solutions fitting the long-period component of the lightcurve satisfactorily. The second solution agrees with the radar solution within the error bars. The parameters of these solutions are summarized in Table 2, with their errors obtained using procedure described in Section 2.5. The LS photometric law was used in the modeling and all sessions of the data were taken as absolutely calibrated (i.e., $P = 1$ in Eq. (6)).

The long-period component of the data together with the synthetic lightcurves of the best-fit solutions are presented in Fig. 9. The regions of plausible values of λ_p and β_p are plotted in Fig. 10 together with the best-fit solutions from this work and from Margot et al. (2002). The solution I from photometry and the radar solution are in a very good agreement. The result presented in Fig. 10 is a nice example of complementarity of the two independent methods; the radar rules out the other solution, while the photometric solution constrain λ_p much better.

3.5. (66063) 1998 RO$_1$

The binary nature of this near-Earth asteroid was discovered from photometric data and confirmed by radar by Pravec et al. (2003a). Pravec et al. (2006) investigated its lightcurve data taken in four apparitions—in September 2001, 2002, 2003 and 2004. They derived $P_{\mathrm{orb}}^{\mathrm{syn}} = 14.53 \pm 0.02$ h and 14.55 ± 0.01 h from the

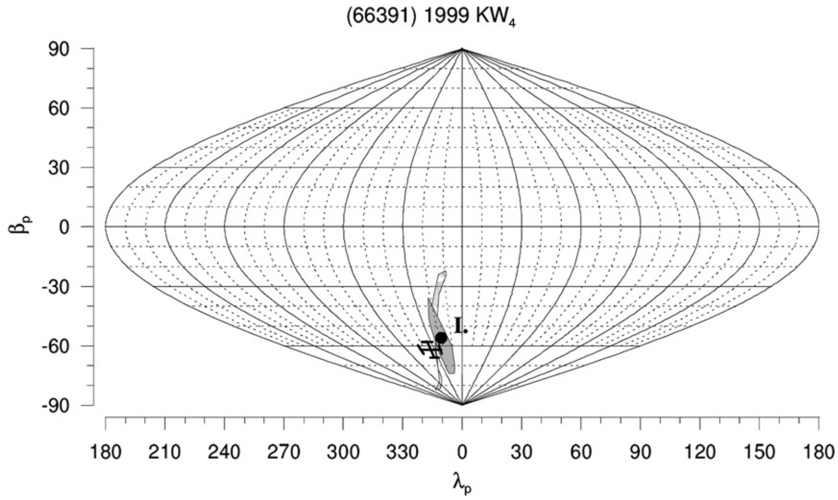

Fig. 8. Ranges of plausible poles of the mutual orbit of (66391) 1999 KW_4 for apparitions 2000 (light gray region) and 2001 (dark grey region) in ecliptic coordinates. The best-fit solutions from this work (apparition 2001; filled circle) and from Ostro et al. (2006) (cross, with 3-σ formal error bars indicated) are denoted as well.

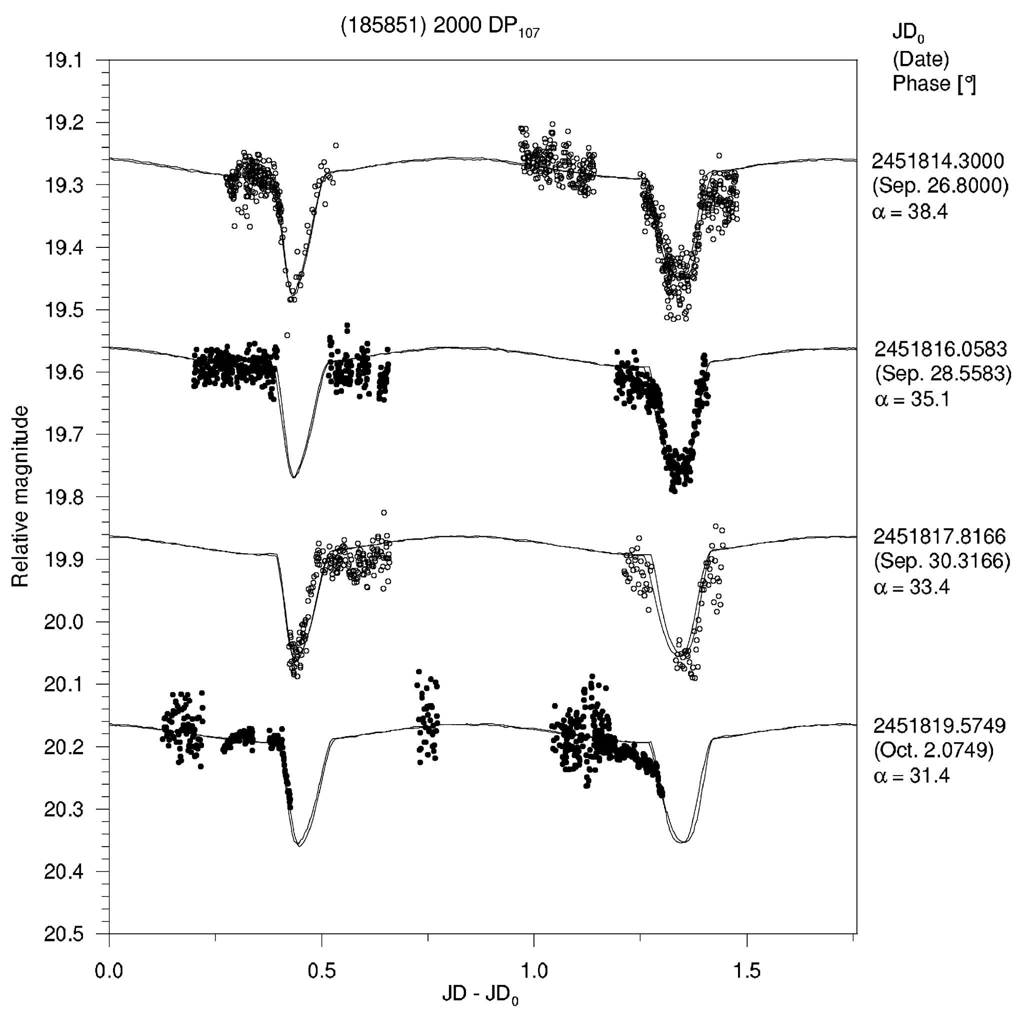

Fig. 9. Long-period lightcurve component of (185851) 2000 DP_{107}. The observational data (points) are plotted together with the synthetic lightcurves for the best-fit solutions (curves). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.3 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD_0) are listed in the right column for each curve, as well as the phase angle α.

2003 and 2004 data, respectively. The rough estimate of the orbital period from the less abundant data of 2002 was $P_{\mathrm{orb}}^{\mathrm{syn}} = 14.45 \pm 0.05$ h. In the 2001 data the secondary component was not resolved due to very short time span of data (3.5 h). From the depth of the secondary event detected in 2002 they estimated $D_2/D_1 = 0.48 \pm 0.03$.

Scheirich (2008) fitted 2003 and 2004 data without D_2/D_1 held fixed, and found two solutions for 2003 data and one solution for

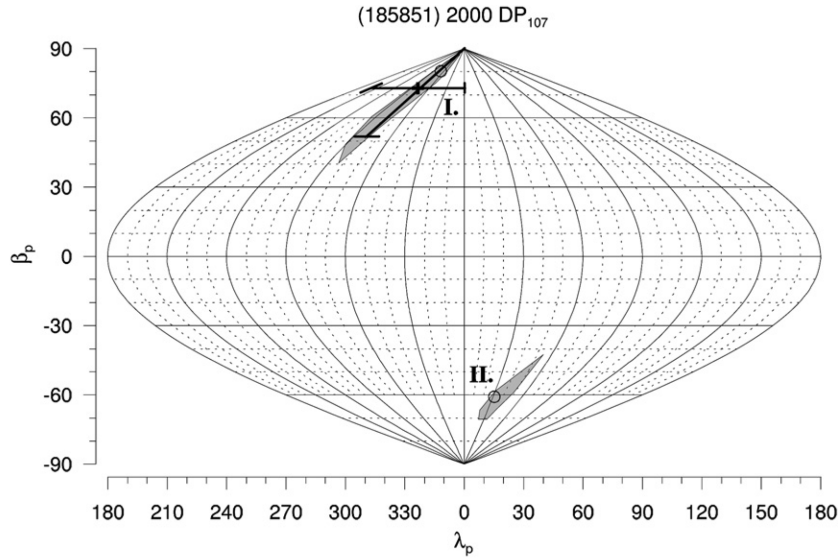

Fig. 10. Range of plausible poles of the mutual orbit of (185851) 2000 DP_{107} in ecliptic coordinates. The best-fit solutions from this work (open circles) and from Margot et al. (2002) (cross, with 3-σ formal error bars indicated) are denoted as well.

2004 data. The first solution for 2003 and the solution for 2004 are consistent with the two solutions from this work, except for the size of the secondary (see below).

We reanalysed the data by Pravec et al. (2006) taken during 2003-09-16.8 to 09-25.9 and during 2004-09-10.2 to 09-21.3 using the model of two-axis ellipsoids on the eccentric orbit described in Section 2.1. While the secondary events in these two apparitions appeared to be nearly but not entirely total due to variation on their bottoms (see Pravec et al., 2006), we choose not to fit D_2/D_1, but to use a fixed value of 0.48 derived by Pravec et al. (2006) from 2002 apparition, instead. We found unique solutions for both 2003 and 2004 apparition, fitting the long-period component of the lightcurve satisfactorily. The parameters of these solutions are summarized in Table 2, with their errors obtained using procedure described in Section 2.5 (the errors for D_2/D_1 are taken from Pravec et al., 2006).

The LS photometric law was used in the modeling and all sessions of the data were taken as absolutely calibrated (i.e., $P = 1$ in Eq. (6)). For comparison of different scattering laws, the minimization was carried out using the Hapke's law for macroscopically rough surface for 2004 data also, but the obtained parameters did not differ significantly.[16]

The long-period components of 2003 and 2004 data together with the synthetic lightcurves of the best-fit solutions are presented in Figs. 12 and 13, respectively. The regions of plausible values of λ_p and β_p are plotted in Fig. 14 together with the best-fit solutions from this work.

The best-fit solutions from both apparitions are near the edges of the regions of plausible poles. The asymmetry is because the eclipse/occultation geometry moves away from nearly total quite rapidly with changing λ_p, β_p in certain directions, while in other directions the geometry remains nearly total in a broader range of the pole positions.

The variations on the bottoms of secondary events of 2003 apparition, which repeat in all events, as is clearly seen in Fig. 6 from Pravec et al. (2006), is not explained by the synthetic lightcurves of the best-fit solution. We suppose that it is caused by a specific shape of the secondary (i.e., that we see a part of rotational lightcurve of the secondary, visible during partial occultation/eclipse), not described by the approximation with ellipsoidal shape.

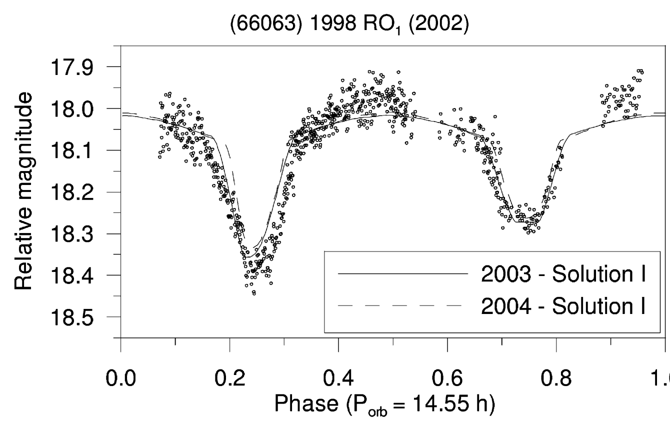

Fig. 11. Composite long-period lightcurve component of (66063) 1998 RO_1 (2002 apparition). The observational data (points) are plotted together with the synthetic lightcurves (curves) constructed using the parameters of best-fit solutions from 2003 and 2004 apparition. The LS photometric law was used.

Due to insufficient coverage of the 2002 data (taken during 2002-09-13.0 to 09-16.0, see Pravec et al., 2006, for details), we were unable to run the full analysis of them and they were used for checking consistency with the models from 2003 and 2004 data only. Fig. 11 shows the long-period component of the data together with the synthetic lightcurves of the best-fit solutions from 2003 and 2004 apparition (the secondary's argument of mean length L_0 was fitted to match the phase of the events).

4. Concluding remarks

We found a remarkable agreement of results from the two different techniques—photometry and radar observations, for two binary near-Earth asteroids, and we estimated parameters for other three binaries.

In Fig. 15, we plotted derived bulk densities and their uncertainties. Formal best-fit values of densities are close to lower limits of the error bars. Upper limits of the error bars are large—this is because flattenings of primaries are poorly constrained by observed data for short observing arcs, and they propagate to the uncertainties of the bulk densities.

[16] We used parameters of Hapke's law which Bowell et al. (1989) derived for S-class asteroid, 82 Alkmene. See their Table IV.

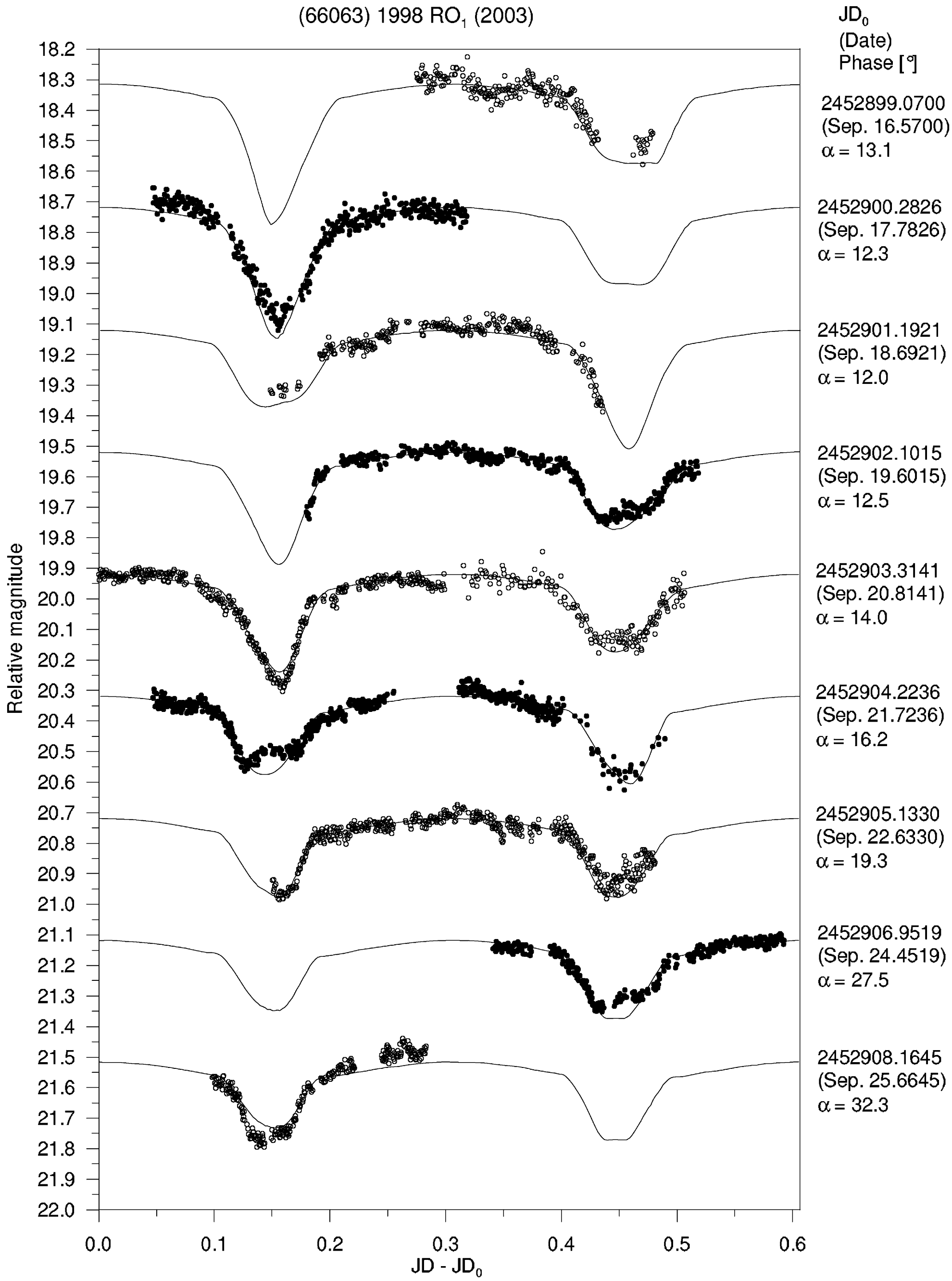

Fig. 12. Long-period lightcurve component of (66063) 1998 RO_1 (2003 apparition). The observational data (points) are plotted together with the synthetic lightcurve for the best-fit solution (curve). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.4 mag for clarity, and different symbols are used for them to avoid confusion. The offsets in date (JD_0) are listed in the right column for each curve, as well as the phase angle α. On the third and sixth curve from the top the minima are shown in an order opposite to other curves.

We again point out that bulk densities estimated from photometric data should be used with caution, and few conclusions on structure and composition of the bodies can be drawn from them.

Acknowledgments

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Appendix A. Explanation of symbols used in text

The symbols used for a specific purpose in a single section only are omitted.

a	semimajor axis of the mutual orbit
A_1, C_1	equatorial and polar semiaxes of the primary represented as an oblate spheroid
A_2, C_2	the longest and the shortest semiaxes of the secondary represented as a prolate spheroid
$\vec{c}^0$	unit vector pointing from primary's to secondary's center
D_1, D_2	mean projected diameters of the primary and the secondary (or, more generally, equivalent diameters when the shape is not specified)
e, ω	eccentricity and argument of pericenter of the mutual orbit

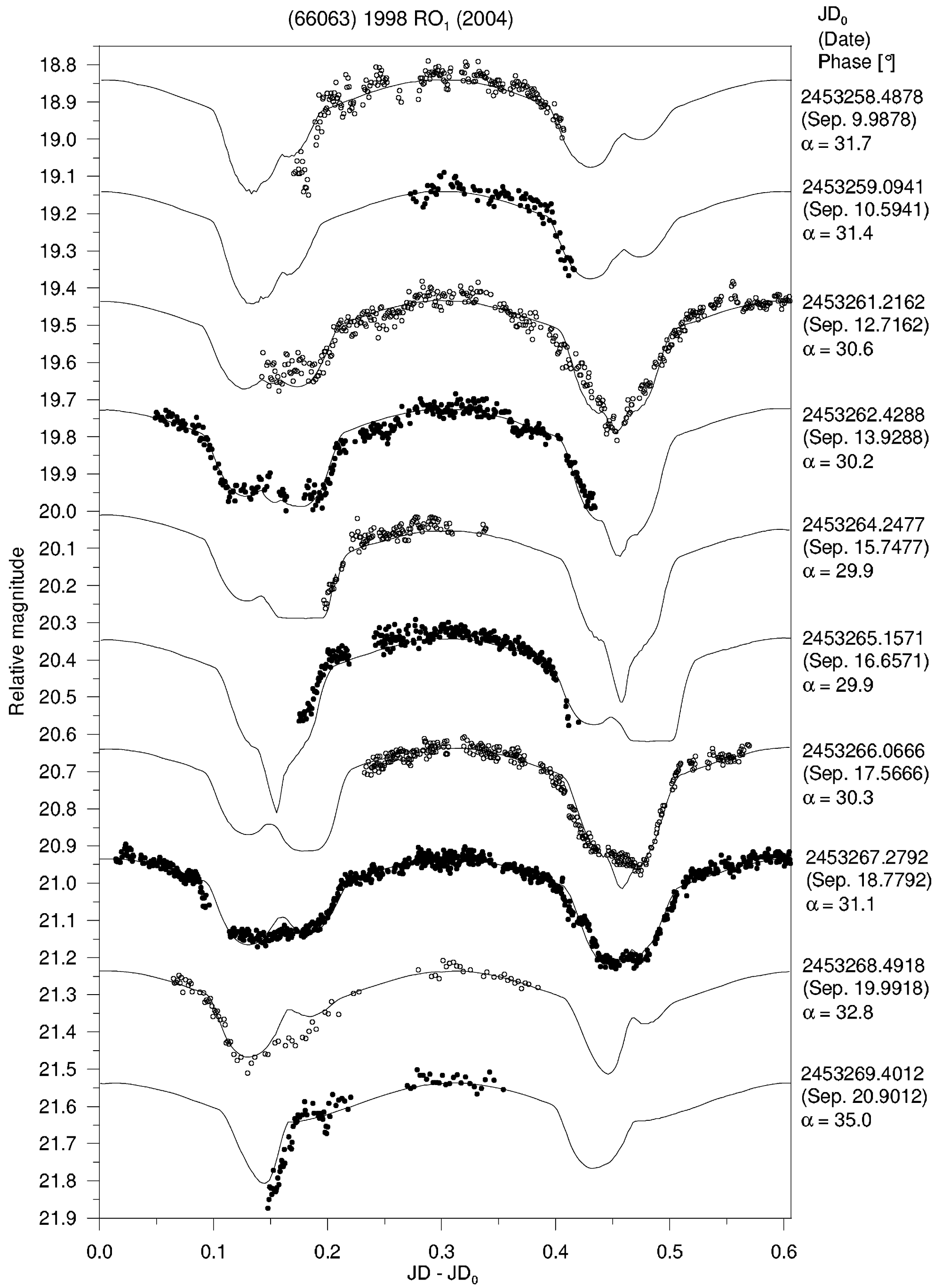

Fig. 13. Long-period lightcurve component of (66063) 1998 RO_1 (2004 apparition). The observational data (points) are plotted together with the synthetic lightcurve for the best-fit solution (curve). The LS photometric law was used. The lightcurve sequences from different dates are vertically offset by 0.3 mag for clarity, and different symbols are used for them to avoid confusion. On the second, third, fifth, sixth and seventh curve from the top the minima are shown in an order opposite to other curves.

L	$(= \omega + M)$ argument of mean length of the secondary
L'	time-increasing value of L (not restricted to interval $[0, 2\pi)$ only)
M	mean anomaly of the secondary
n	mean motion of the secondary
$\vec{n}^0$	unit vector normal to the secondary's orbit
P_1, P_2	periods of components of the lightcurve (the nature of components is not specified)
$P_{\mathrm{orb}}^{\mathrm{syn}}, P_{\mathrm{orb}}^{\mathrm{sid}}$	synodic and sidereal orbital period of the system
R_1, R_2	$= D_1/2, D_2/2$
$\vec{s}^0, \vec{h}^0$	unit vectors pointing from the asteroid to the Sun and to the Earth
T	time epoch
α	phase angle
β_s	angle between $\vec{s}^0$ (or $\vec{h}^0$) and $\vec{n}^0$
δ_1	angle between $\vec{c}^0$ and $\vec{s}^0$ (or $\vec{h}^0$) for the first and the fourth contact
δ_2	angle between $\vec{c}^0$ and $\vec{s}^0$ (or $\vec{h}^0$) for the second and the third contact
$\lambda_{\mathrm{p}}, \beta_{\mathrm{p}}$	pole coordinates of the mutual orbit in ecliptic frame
ρ	bulk density of components of binary system

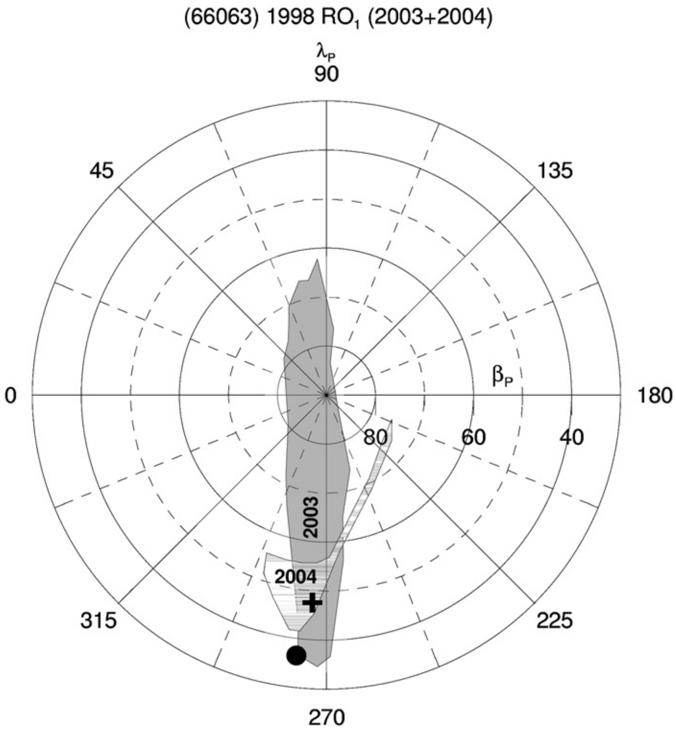

Fig. 14. Ranges of plausible poles of the mutual orbit of (66063) 1998 RO_1 for apparitions 2003 and 2004 in ecliptic coordinates (view from the north ecliptic pole). The best-fit solutions (filled circle for 2003 and cross for 2004) are denoted as well.

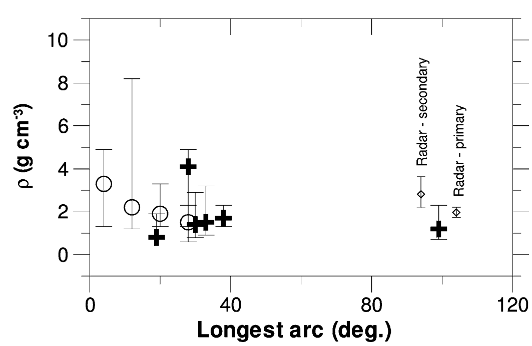

Fig. 15. Uncertainties of densities of the systems modeled in this work (crosses) and in Scheirich (2008) (open circles). The abscissa represents the longest of the arcs spanned by the asteroid with respect to the Earth and to the Sun. The rightmost point represents (66391) 1999 KW_4. Two diamonds next to it (horizontally offset) indicate the best-fit solutions with standard errors of the bulk density derived for the primary and the secondary by Ostro et al. (2006).

Appendix B. Radius vectors of secondary body in apparent contacts

The algorithms described in this work need as input unit vectors in the primary-to-secondary direction in instants of time when certain contacts (see Section 2) occur.

It is assumed in the derivation that the geometry with respect to Earth and Sun, and the pole and the semi-major axis of the mutual orbit are given, as well as the size ratio of the components, and that their shapes are spherical.

For simplicity, only the case of circular orbit is described here. Define a unit vector $\vec{c}^0$ pointing from the primary's to the secondary's center at the time of some of the contacts for an eclipse of the primary. Let R_1 and R_2 be radii for primary and secondary, respectively, and r be a radius of the secondary's orbit around the

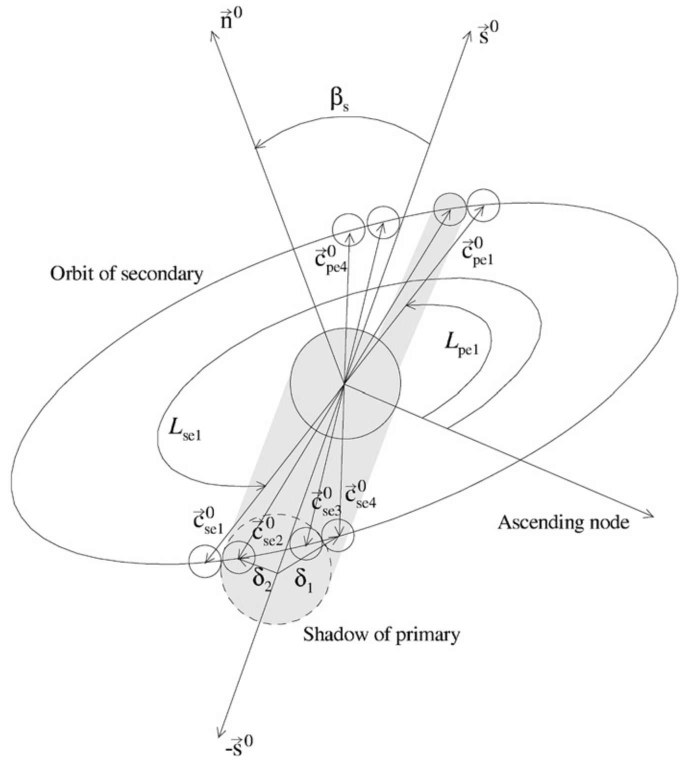

Fig. 16. Vectors and angles used in the derivation of the eclipse contacts: $\vec{c}^0_{\rm se1}$ to $\vec{c}^0_{\rm se4}$ are unit vectors pointing from the primary's to the secondary's center at the first to the fourth contact during secondary's eclipse, $\vec{n}^0$ is the normal to the mutual orbit plane, $\vec{s}^0$ is a vector pointing in the direction to Sun, δ_1 is an angle between $\vec{c}^0_{\rm se1}$ and $-\vec{s}^0$ ($\vec{c}^0_{\rm se4}$ and $-\vec{s}^0$), δ_2 is an angle between $\vec{c}^0_{\rm se2}$ and $-\vec{s}^0$ ($\vec{c}^0_{\rm se3}$ and $-\vec{s}^0$), and $\beta_{\rm s}$ is an angle between $\vec{n}^0$ and $\vec{s}^0$. $L_{\rm pe1}$ and $L_{\rm se1}$ are mean lengths of secondary in the first contacts of secondary's shadow with the primary and the primary's shadow with the secondary, respectively. The shadows of the primary and the secondary are indicated as gray area.

primary. Denote δ_1 an angle between $\vec{c}^0$ for the first (or the fourth) contact and the unit vector in the direction to the Sun, $\vec{s}^0$.[17] Similarly, δ_2 is an analogous angle for the second and the third contact (see Fig. 16).

The vector $\vec{c}^0$ for the first and the fourth contact is then defined by a set of equations

$$\vec{c}^0 \cdot \vec{s}^0 = \cos\delta_1, \tag{B.1}$$

$$\vec{c}^0 \cdot \vec{n}^0 = 0, \tag{B.2}$$

where $\vec{n}^0$ is a unit vector normal to the secondary's orbit.

Solving Eqs. (B.1), (B.2) and $|\vec{c}^0| = 1$, we get two solutions for $\vec{c}^0$:

$$\vec{c}^0 = \frac{\alpha\vec{a} + \vec{b}}{a^2}, \tag{B.3}$$

where

$$\vec{a} = \vec{n}^0 \times \vec{s}^0, \tag{B.4}$$

$$\vec{b} = \vec{a} \times \left(\cos\delta_1 \vec{n}^0\right), \tag{B.5}$$

and

$$\alpha = \pm\sqrt{a^2 - b^2/a^2}. \tag{B.6}$$

We denote the two solutions as $\vec{c}^0_{\rm pe1}$ and $\vec{c}^0_{\rm pe4}$ for the first and the fourth contact of the primary eclipse, respectively.

[17] Note that the vector $\vec{s}^0$ do not lie in the secondary's orbital plane, in general.

Positions of the secondary at the second ($\vec{c}^{\,0}_{\rm pe2}$) and the third ($\vec{c}^{\,0}_{\rm pe3}$) contacts could be found in analogous way, using angle δ_2 instead of δ_1.

Substituting $\vec{s}^{\,0}$ with $-\vec{s}^{\,0}$ in Eq. (B.4), we obtain positions of the secondary in four contacts of eclipse of secondary, $\vec{c}^{\,0}_{\rm se1}$ to $\vec{c}^{\,0}_{\rm se4}$. Finally, substituting $\vec{s}^{\,0}$ with the vectors pointing in the direction to Earth, $\vec{h}^0$, and in the opposite sense, $-\vec{h}^0$, we obtain contacts of secondary's transit in front of the primary, $\vec{c}^{\,0}_{\rm po1}$ to $\vec{c}^{\,0}_{\rm po4}$, and of secondary occultation, $\vec{c}^{\,0}_{\rm so1}$ to $\vec{c}^{\,0}_{\rm so4}$, respectively.

We assume in the above derivations that the geometry is constant during the whole event (i.e. the vectors $\vec{s}^{\,0}$ and $\vec{h}^0$ are fixed). If the geometry is changing rapidly, the vectors $\vec{c}^{\,0}_i$ have to be searched iteratively.

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